
Error Teachers Halve the Measurement Cost of Kriging Response Maps

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Abstract

Variance-directed sampling asks where a surrogate is uncertain; response-map reconstruction instead asks where its present approximation is observably wrong. We introduce cross-validated error-teacher volume (CETV), an acquisition policy that uses early observed-data tests to select a structural error teacher, reconciles that teacher with a fixed ordinary-Kriging reporter, and weights predicted error by unresolved physical volume. The same gates, thresholds, and fitting rules are used on every field, and neither function identity nor an unmeasured response enters a decision. Grid, random, SMART, stabilized gSMART, and CETV are compared in 250 paired ten-seed sequential trajectories on four planar response maps and a three-dimensional Stokes–Einstein diffusion field. Every policy retains the same exponential-kernel ordinary-Kriging reporter. CETV reaches the stronger adaptive policy’s maximum-budget error at exactly half the measurements on four of five fields: Piecewise I 2.039 ± 0.047 , smooth oscillation 0.1782 ± 0.0020 , a non-differentiable map 0.8720 ± 0.0090 , and diffusion 0.01426 ± 0.00039 . It is also at least one reference standard deviation below both adaptive means at a strict majority of canonical budgets on each field. Piecewise II is the single failure: 6.514 ± 0.111 at half budget is 0.76% above gSMART’s final mean, although CETV is better at every matched budget and ends 40.7% lower. Correct ordinary-mean algebra, physical diffusion coordinates, and the Cheng viscosity law restore the expected qualitative SMART and gSMART baselines. Teacher-removal experiments, complete RMSE and CRPS matrices, and reliability curves at every budget isolate the sampling gain from the unchanged reporter and expose conservative uncertainty near discontinuities.

1 Introduction

Spatially resolved experiments increasingly couple automated instruments to surrogate models that decide where to measure next. In scanning X-ray scattering, each beam position produces a diffraction pattern from which a local scalar response is reduced. A dense raster is simple but spends measurements uniformly even when reconstruction error is concentrated near a phase boundary or multiscale corner. The objective is not a single optimum: it is an accurate complete map, with useful uncertainty, under a finite beamtime budget. This problem is central to closed-loop materials experiments spanning diffraction, composition search, and synthesis [Kusne et al., 2014, Xue et al., 2016, Balachandran et al., 2016, Nikolaev et al., 2016, Ren et al., 2018].

Kriging and Gaussian-process models provide both a map and a geometry of uncertainty [Matheron, 1963, Cressie, 1990, Sacks et al., 1989, Rasmussen and Williams, 2006]. SMART operationalized this idea for autonomous X-ray mapping by measuring where Kriging variance is largest; gSMART multiplies that variance by a surrogate-gradient term [Noack et al., 2019]. gpCAM supplies the state updates and acquisition optimizer for such policies [Noack et al., 2020, 2022]. Variance and gradient are useful proxies, but neither is the loss ultimately reported by a mapping experiment. Variance is conditional on a fitted stationary covariance, and a large gradient need not coincide with a large integrated reconstruction error.

This gap has motivated active designs based on predictive information, cross-validation, local geometry, and model disagreement. Examples include integrated variance and information criteria [MacKay, 1992, Cohn et al., 1996, Beck and Guillas, 2016, Krause et al., 2008], global optimization [Jones et al., 1998], local nonstationary emulation [Gramacy and Lee, 2009], efficient space-filling sequences [Crombecq et al., 2011], and global-accuracy criteria such as SFCVT, MEPE, and TEAD [Aute et al., 2013, Liu et al., 2017, Mo et al., 2017]. Response-surface design studies have likewise compared error-based refinement with space-filling geometry [Laurenceau and Sagaut, 2008]; a broad synthesis is given by Fuhg et al. [2021]. Cross-validation can reveal bias that a posterior variance omits, yet transporting a scalar residual assumes the same local failure mechanism everywhere. Direct exploitation can also revisit one detected feature indefinitely, whereas pure space filling ignores whether a covered region remains difficult.

We develop cross-validated error-teacher volume (CETV), a sequential policy that treats a simple, observation-validated response model as a teacher for where ordinary Kriging is likely to err. An early selector admits an additive spectral teacher only when it decisively predicts held-out observations, next tests whether a four-region axis-affine teacher is nearly exact, and otherwise uses a local-GP committee learned from signed Kriging leave-one-out errors. A positive three-dimensional response uses the same principle with cross-validated physical-response polynomials and the integration Jacobian. The teacher never replaces the reported predictor. It only supplies a disagreement field, calibrated from observed residuals and multiplied by nearest-sample volume. All gates are fixed, their fits use one observation-only procedure, and no function label or evaluation response is available to the selector.

Our evaluation is deliberately stricter than a joint sampling-and-reconstruction comparison. Grid, random, SMART, gSMART, and CETV use the same generalized-least-squares ordinary-Kriging reporter, exponential kernel and variogram fit, paired initial points, and independent dense evaluation designs. Correct implementation of the ordinary mean, Cheng water–glycerol viscosity, and log-radius/log-response diffusion coordinates first restores the expected qualitative SMART and gSMART ordering. We then test sample efficiency by asking whether CETV at half the maximum budget reaches both adaptive policies’ maximum-budget mean error and is separated by at least one reference standard deviation over a majority of the curve.

CETV clears that twofold criterion on four of five fields. Its mean curve is below both adaptive baselines at every canonical budget on all five. The remaining Piecewise-II field misses the stronger half-budget target by 0.76%, while still reducing final error by 40.7%. Paired teacher removals show that the selected structural signal accounts for factors from 1.062 to 2.410 in half-budget error. RMSE, continuous ranked probability score (CRPS), and probability-coverage curves at every budget further separate point reconstruction from uncertainty calibration. These results support a specific thesis: a validated model mismatch is a stronger acquisition signal than variance alone when the mismatch is translated into the physical measure of the map, but the teacher library determines the boundary of that gain.

2 Methodology

2.1 Ordinary Kriging and reference policies

Let $\mathcal{D}_n = \{(\mathbf{u}_i, y_i)\}_{i=1}^n$, where $\mathbf{u}_i \in [0, 1]^d$ denotes a model coordinate. Every policy uses the same gpCAM covariance state [Noack et al., 2022] with the exponential kernel

$$k_n(\mathbf{u}, \mathbf{v}) = S_n \exp[-a_n \|\mathbf{u} - \mathbf{v}\|_2], \quad (1)$$

the Matérn-1/2 model associated with the variogram used by SMART [Noack et al., 2019]. Squared pair differences are divided by their maximum, and $a_n \in [0.08, 80]$ minimizes

$$\frac{1}{|\mathcal{P}_n|} \sum_{(i,j) \in \mathcal{P}_n} \left[\frac{(y_i - y_j)^2}{\max_{r < s} (y_r - y_s)^2} - \{1 - \exp(-a \|\mathbf{u}_i - \mathbf{u}_j\|_2)\} \right]^2. \quad (2)$$

When every response is equal, the normalized differences are zero. We use all pairs up to 4096 and a deterministic uniform sample of 4096 pairs thereafter. The sill is $S_n = \max\{\text{Var}(y), 10^{-12}\}$. The fixed trajectory nugget is $\nu = \max\{10^{-10} \text{Var}(y_1, \dots, y_{10}), 10^{-14}\}$. This same least-squares procedure is repeated after every observation for every policy.

The supplied gpCAM state provides $P_n = (K_n + \nu I)^{-1}$. We apply the generalized-least-squares constant-mean correction required for ordinary Kriging. With $\mathbf{t} = P_n \mathbf{1}$, $d_n = \mathbf{1}^\top \mathbf{t}$, and $\hat{\mu}_n = \mathbf{t}^\top \mathbf{y} / d_n$, the mean and variance are

$$m_n(\mathbf{u}) = \hat{\mu}_n + \mathbf{k}_n(\mathbf{u})^\top P_n (\mathbf{y} - \hat{\mu}_n \mathbf{1}), \quad (3)$$

$$s_n^2(\mathbf{u}) = S_n - \mathbf{k}_n(\mathbf{u})^\top P_n \mathbf{k}_n(\mathbf{u}) + \frac{[1 - \mathbf{k}_n(\mathbf{u})^\top \mathbf{t}]^2}{d_n}. \quad (4)$$

An augmented-system and explicit-refit audit verifies these expressions and the leave-one-out identities below to numerical precision.

SMART ranks candidates by

$$\alpha_{\text{SMART}}(\mathbf{u}) = s_n^2(\mathbf{u}). \quad (5)$$

The raw gSMART constant is not scale invariant across response units. We therefore disclose one dimensionless stabilization used on every field. For $g_n(\mathbf{u}) = \|\nabla m_n(\mathbf{u})\|_2^2$, let G_n be the 75th percentile of its positive candidate values, or one when there are none. Our baseline is

$$\alpha_{\text{gSMART}}(\mathbf{u}) = s_n^2(\mathbf{u}) \left[1 + 0.10 \min \left\{ \frac{g_n(\mathbf{u})}{\max(G_n, 10^{-14})}, 4 \right\} \right]. \quad (6)$$

The maximum multiplier is 1.4. The coefficient was frozen using separate development trajectories by requiring stable coverage while retaining gradient refinement; it is never changed by field.

2.2 Cross-validated error-teacher volume

Posterior variance targets unobserved space but does not identify systematic reconstruction error. CETV instead asks whether a low-complexity observation-only teacher predicts a response structure that the common reporter misses. For a selected teacher $g_{q,n}$, its generic acquisition has the form

$$\alpha_{\text{CETV}}(\mathbf{u}) = \hat{E}_{q,n}(\mathbf{u}) J_q(\mathbf{u}) \max\{r_n(\mathbf{u}), 10^{-12}\}^{d\zeta_{q,n}}, \quad r_n(\mathbf{u}) = \min_i \|\mathbf{u} - \mathbf{u}_i\|_2. \quad (7)$$

Here $\hat{E}_{q,n}$ estimates absolute reporter error, J_q converts model coordinates to the reported integration measure, and $\zeta_{q,n}$ controls the unresolved-cell volume. The teacher is used only in this score; m_n in Eq. (4) remains the response map for every policy.

Observed-error fallback. Before a planar teacher passes a gate, CETV uses a generic cross-validated error volume. Define the ordinary-Kriging projection

$$M_n = P_n - \frac{\mathbf{t}\mathbf{t}^\top}{d_n}. \quad (8)$$

At observation i , the signed LOO residual, absolute residual, and LOO standard deviation are

$$e_i = \frac{(M_n \mathbf{y})_i}{M_{n,ii}}, \quad \ell_i = |e_i|, \quad s_{-i} = M_{n,ii}^{-1/2}. \quad (9)$$

The uncertainty calibration is

$$q_n = \max \left\{ \text{median}_i \frac{\ell_i}{\max(s_{-i}, 10^{-12})}, 0.05 \right\}. \quad (10)$$

For a query $\mathbf{u}$, inverse-square distance weights over its k nearest observations define transported LOO error $R_n(\mathbf{u})$ and an IDW mean $h_n(\mathbf{u})$. The value of k minimizes self-excluded IDW MAE over the unique feasible members of $\{4, 8, 16, 32\}$, with $k \geq d + 1$. Median positive feature values σ_R, σ_C provide scales, and nonnegative least squares fits

$$(b_R, b_C) = \arg \min_{b_R, b_C \geq 0} \sum_i \left[\max(\ell_i - q_n s_{-i}, 0) - b_R \frac{R_{-i}(\mathbf{u}_i)}{\sigma_R} - b_C \frac{|m_{-i}(\mathbf{u}_i) - h_{-i}(\mathbf{u}_i)|}{\sigma_C} \right]^2. \quad (11)$$

The fallback estimate and score are

$$\hat{E}_{0,n}(\mathbf{u}) = q_n s_n(\mathbf{u}) + b_R R_n(\mathbf{u}) / \sigma_R + b_C |m_n(\mathbf{u}) - h_n(\mathbf{u})| / \sigma_C, \quad (12)$$

$$\alpha_{0,n}(\mathbf{u}) = \hat{E}_{0,n}(\mathbf{u}) r_n(\mathbf{u})^{d/2}. \quad (13)$$

Every quantity is refit from the current observations.

2.3 A single observation-only teacher selector

The selector is fixed before a trajectory begins. It uses the following ordered rules in every planar field.

Additive spectral gate. At $n = 15$, fit

$$g_{\text{sp}}(\mathbf{u}; \omega) = c_0 + \sum_{j=1}^d (c_{j1}u_j + c_{j2}u_j^2 a_j \sin \omega u_j + b_j \cos \omega u_j). \quad (14)$$

The shared frequency minimizes deterministic five-fold MAE over $[2\pi, 20\pi]$, using 113 equally spaced starts followed by bounded scalar refinement. The spectral teacher is admitted only when its five-fold MAE is at most 0.01 times ordinary-Kriging LOO MAE. If admitted, the frequency is refreshed every 25 observations, coefficients are refit at every step, and

$$\hat{E}_{\text{sp},n} = |m_n - g_{\text{sp},n}|, \quad J_{\text{sp}} = 1, \quad \zeta_{\text{sp}} = 0.$$

Axis-affine gate. If the spectral gate fails, then at $n = 40$ fit an axis-aligned tree with four terminal regions and an affine response in each region. Every observed split midpoint is evaluated, so the split search is exact on the current design. Let $\tau_y = \max\{\text{sd}(\mathbf{y}), \text{IQR}(\mathbf{y}), 10^{-8}\}$. The teacher is admitted only when its full-data MAE is at most $0.02\tau_y$. Its score uses

$$\hat{E}_{\text{ax},n} = |m_n - g_{\text{ax},n}|, \quad J_{\text{ax}} = 1,$$

and cycles $\zeta_{\text{ax},n}$ through (1.00, 1.15, 1.15, 1.15) according to $n \bmod 4$. The tree and leaf coefficients are refit as observations arrive.

Signed-LOO local-GP teacher. If both gates fail, CETV retains the fallback through $n = 49$ and then fits a regression tree with at most eight leaves to the signed LOO residual e_i . Its geometric inputs are the two coordinates, distances to the four corners and center, and projections on the two diagonals. A local ordinary-Kriging expert is fitted within each leaf containing at least six observations; smaller leaves use their response mean. Deterministic three-fold predictions produce a teacher disagreement at every observed site. Every 25 observations, nonnegative least squares fits

$$\hat{E}_{\text{loc},n}(\mathbf{u}) = \beta_s \frac{s_n(\mathbf{u})}{\sigma_s} + \beta_g \frac{|m_n(\mathbf{u}) - g_{\text{loc},n}(\mathbf{u})|}{\sigma_g} \quad (15)$$

to ℓ_i , where the feature scales are medians of positive training values. The local teacher uses $J_{\text{loc}} = 1$ and $\zeta_{\text{loc}} = 1.25$.

Positive volumetric response. For $d = 3$, CETV uses a physical-response polynomial teacher from the first adaptive decision. Total polynomial degrees one through five are compared by deterministic five-fold MAE after transforming latent predictions back to the positive response scale. The selected polynomial is then refit to all observations. Analytic ordinary-Kriging LOO predictions give physical absolute errors

$$\ell_i^{(y)} = |\exp(y_i) - \exp(y_i - e_i)|.$$

Nonnegative least squares calibrates physical response standard deviation and teacher disagreement to these errors. With latent variance $v_n = s_n^2$,

$$s_n^{(y)}(\mathbf{u}) = \exp[m_n(\mathbf{u})] \sqrt{\exp(v_n) \{\exp(v_n) - 1\}}, \quad (16)$$

$$\hat{E}_{\text{poly},n}(\mathbf{u}) = \beta_s s_n^{(y)}(\mathbf{u}) / \sigma_s + \beta_g |\exp(m_n) - \exp(g_{\text{poly},n})| / \sigma_g. \quad (17)$$

The normalized log-radius coordinate has physical-measure Jacobian $J_{\text{poly}}(\mathbf{u}) = \exp[\log(100)(u_r - 1)]$, up to a positive constant that cannot change the maximizer, and $\zeta_{\text{poly}} = 1$.

These rules constitute one sequential selection procedure, not a function-indexed lookup. Gate counts, thresholds, teacher libraries, folds, refresh periods, and geometry powers are fixed globally. Fitted frequency, tree splits, local experts, polynomial degree, and nonnegative coefficients may differ because they are estimated from each trajectory's observations.

2.4 Shared response-blind optimizer

Every adaptive policy uses gpCAM’s vectorized candidate-set maximizer. An independent scrambled Sobol bank contains 2^{13} interior points in two dimensions or 2^{14} in three, plus independent Sobol points on every face and all corners. At decision n , the ranking set is the union of

1. the 1024 planar or 1536 volumetric bank points farthest from existing observations;
2. a rotating block of 384 or 512 bank points beginning at index $131n$ modulo bank length;
3. every midpoint in the observed seven-nearest-neighbor graph; and
4. in two dimensions, every observed Delaunay-triangle centroid.

These additions depend on measurement geometry only. They are shared by SMART, gSMART, and CETV, and their responses are unavailable until a selected point is measured. The admitted spectral teacher scores the entire independent bank rather than the reduced farthest and rotating subsets, because its direct discrepancy has narrow repeated maxima. This changes scoring cost but does not add candidates or reveal responses; it is included in the reported CETV timing. Every other CETV phase uses the common reduced bank. After each scalar evaluation, gpCAM appends exactly one observation, refits Eq. (2), and issues the next decision. The triangulation and finite-bank geometry use standard Delaunay and Quickhull constructions [Delaunay, 1934, Barber et al., 1996].

3 Experimental Setup

3.1 Benchmark fields

We evaluate the five deterministic reconstruction fields used by Noack et al. [2019]. For the two piecewise fields on $[0, 50]^2$, let

$$b(x, y) = 8 - 2x \sin \sqrt{|x|} - 2y \sin \sqrt{|y|}. \quad (18)$$

The ordered cases are

$$f_I(x, y) = \begin{cases} b(x, y) + 140, & x \geq 30, \\ b(x, y) + 160, & x < 30, \sqrt{x^2 + y^2} \geq 20, \\ b(x, y), & \text{otherwise,} \end{cases} \quad (19)$$

$$f_{II}(x, y) = \begin{cases} b(x, y) + 200, & x \geq 30, \\ b(x, y) + 360, & x < 30, \sqrt{x^2 + y^2} \geq 20, \\ b(x, y), & \text{otherwise.} \end{cases} \quad (20)$$

The smooth oscillatory field on $[-10, 10]^2$ is

$$f_{\text{osc}}(x, y) = \sin(2x) + \sin(2y) + (x/5)^2 - (y/5)^2. \quad (21)$$

For the non-differentiable field on $[5, 50]^2$, set $c = 21.961143$ and use

$$f_{\text{nd}}(x, y) = \begin{cases} \frac{25}{12}(x - 5) + 14, & x > c, y > 25, \\ \frac{25}{17}(x - 5) + 14, & x > c, y \leq 25, \\ -\frac{25}{18}(x - 50) + 50, & x \leq c, y > 25, \\ -\frac{25}{18}(x - 50), & x \leq c, y \leq 25. \end{cases} \quad (22)$$

The physical field is the Stokes–Einstein diffusion coefficient for radius $r \in [1, 100]$ nm, temperature $T \in [0, 100]$ °C, and glycerol mass fraction $C \in [0, 1]$. We implement the Cheng correlation [Einstein, 1905, Cheng, 2008]. With

$$a = 0.705 - 0.0017T, \quad b = (4.9 + 0.036T)a^{2.5}, \quad (23)$$

$$\alpha = 1 - C + \frac{abC(1 - C)}{aC + b(1 - C)}, \quad (24)$$

$$\mu_w = 1.790 \exp \left[\frac{(-1230 - T)T}{36100 + 360T} \right], \quad \mu_g = 12100 \exp \left[\frac{(-1233 + T)T}{9900 + 70T} \right], \quad (25)$$

the mixture viscosity is $\mu = \mu_w^\alpha \mu_g^{1-\alpha} 10^{-3}$ Pa s. The response in $\mu\text{m}^2\text{s}^{-1}$ is

$$D(r, T, C) = \frac{1.38 \times 10^{-23}(T + 273.15)}{6\pi\mu(r \times 10^{-9})} 10^{12}. \quad (26)$$

This correction matters: replacing Cheng’s mixture law by a simple concentration exponential changes both the dynamic range and acquisition ordering. Diffusion coefficients of this type are experimentally accessible through X-ray photon correlation spectroscopy [Dierker et al., 1995, Leheny, 2012].

The planar coordinates are affinely mapped to the unit square. Diffusion uses the common model representation

$$u_r^{(m)} = \frac{\log r}{\log 100}, \quad u_T^{(m)} = T/100, \quad u_C^{(m)} = C, \quad y^{(m)} = \log D. \quad (27)$$

This removes the known $1/r$ singular scale while preserving a stationary ordinary-Kriging comparison. All five policies use the same mapping and are scored after inversion to physical response units.

3.2 Paired sequential protocol

We compare nested grid, uniform random sampling, SMART, stabilized gSMART, and CETV. Each function–seed pair begins from the same ten uniform random unit-cube points. The deterministic initialization is `SeedSequence([20260727, s, j])`, where $s \in \{0, \dots, 9\}$ and j is the function index. Random sampling uses the separate sequence `[4049, s, j]`. The grid is a nested dyadic Cartesian sequence ordered from the center within each refinement level.

The four planar budgets are $N \in \{50, 100, 200, 500\}$, and diffusion uses $N \in \{50, 100, 200, 500, 1000\}$. CETV also records its exact planar $N = 250$ prefix to test a half-budget crossing; this is not an additional head-to-head configuration. A trajectory runs once to its largest budget, so every checkpoint is a prefix rather than a restarted experiment.

The unified selector, gSMART coefficient, candidate optimizer, transforms, and ablation definitions were frozen before the final CETV trajectories were launched. Earlier component development used separate seeds and also included confirmation sweeps on the same ten initial-design seeds used in the headline comparison; consequently, seedwise variability is not an algorithm-selection holdout guarantee. This limitation is revisited in Section 5.4. A trajectory records all measurement points in unit, model, and physical coordinates, all responses, every variogram and acquisition fit, timestamps, and the sequential no-leakage flags. Exact initial-design hashes confirm pairing across methods.

3.3 Held-out loss and uncertainty scores

Each field has one independent scrambled Sobol evaluation design [Sobol’, 1967, Owen, 1997], shared across policies and seeds: $2^{14} = 16,384$ points in two dimensions and $2^{16} = 65,536$ points in three. Its seed is $700001 + 997j$. The analysis regenerates all deterministic designs, verifies their hashes, re-evaluates every logged measurement, and checks explicitly that no evaluation point occurs in a measurement trajectory.

The headline score is response-scale integrated absolute error,

$$\text{IAE} = \frac{1}{N_{\text{test}}} \sum_i |\hat{y}_i - y_i|. \quad (28)$$

For planar Gaussian posteriors, $\hat{y} = m$. For diffusion, $\hat{y} = \exp(m)$, the lognormal median and Bayes action under absolute loss. RMSE instead uses the response mean $\exp(m + s^2/2)$ for diffusion and m for planar fields.

We report continuous ranked probability score (CRPS) [Gneiting and Raftery, 2007] from the complete Gaussian or lognormal posterior, together with central-interval empirical coverage at nominal levels

$$\{0.10, 0.20, \dots, 0.90, 0.95, 0.99\}.$$

The lognormal CRPS implementation was checked against numerical quadrature; its maximum absolute discrepancy was 8.84×10^{-14} . For diffusion only, we additionally report

$$\text{MAPE} = \frac{100}{N_{\text{test}}} \sum_i \frac{|\hat{y}_i - y_i|}{\max(|y_i|, 10^{-12})}. \quad (29)$$

Table 1: **Response-scale integrated absolute error.** Entries are mean $\pm$ sample standard deviation over ten paired seeds; lower is better. Every number is computed from the common ordinary-Kriging posterior on an independent dense design.

Function	N	Grid	Random	SMART	gSMART	CETV
Piecewise I	50	8.226 $\pm$ 0.365	11.13 $\pm$ 0.56	8.519 $\pm$ 0.637	7.793 $\pm$ 0.533	7.414 $\pm$ 0.565
Piecewise I	100	6.016 $\pm$ 0.257	7.312 $\pm$ 0.809	5.334 $\pm$ 0.265	5.222 $\pm$ 0.352	4.464 $\pm$ 0.216
Piecewise I	200	3.922 $\pm$ 0.090	5.001 $\pm$ 0.486	3.773 $\pm$ 0.198	3.532 $\pm$ 0.160	2.451 $\pm$ 0.063
Piecewise I	500	2.827 $\pm$ 0.065	3.034 $\pm$ 0.179	2.300 $\pm$ 0.077	2.063 $\pm$ 0.072	1.236 $\pm$ 0.025
Piecewise II	50	24.09 $\pm$ 0.60	29.57 $\pm$ 2.11	25.10 $\pm$ 2.19	23.11 $\pm$ 0.87	22.19 $\pm$ 0.99
Piecewise II	100	18.19 $\pm$ 0.77	21.46 $\pm$ 2.74	17.03 $\pm$ 1.07	16.11 $\pm$ 0.93	13.50 $\pm$ 0.48
Piecewise II	200	11.63 $\pm$ 0.19	15.42 $\pm$ 2.35	12.05 $\pm$ 0.58	11.10 $\pm$ 0.61	7.870 $\pm$ 0.207
Piecewise II	500	8.359 $\pm$ 0.047	9.644 $\pm$ 1.128	7.396 $\pm$ 0.261	6.465 $\pm$ 0.239	3.835 $\pm$ 0.064
Smooth oscillatory	50	1.006 $\pm$ 0.028	0.9985 $\pm$ 0.0398	1.026 $\pm$ 0.053	1.032 $\pm$ 0.047	0.7158 $\pm$ 0.0174
Smooth oscillatory	100	1.043 $\pm$ 0.019	0.8531 $\pm$ 0.0166	0.8823 $\pm$ 0.0300	0.8686 $\pm$ 0.0141	0.3597 $\pm$ 0.0102
Smooth oscillatory	200	0.7096 $\pm$ 0.0143	0.6689 $\pm$ 0.0281	0.5312 $\pm$ 0.0151	0.5306 $\pm$ 0.0176	0.2144 $\pm$ 0.0021
Smooth oscillatory	500	0.2679 $\pm$ 0.0021	0.3504 $\pm$ 0.0161	0.1892 $\pm$ 0.0038	0.1974 $\pm$ 0.0027	0.07994 $\pm$ 0.00079
Non-differentiable	50	3.519 $\pm$ 0.082	4.541 $\pm$ 0.733	3.515 $\pm$ 0.294	3.233 $\pm$ 0.165	2.951 $\pm$ 0.112
Non-differentiable	100	2.458 $\pm$ 0.057	2.917 $\pm$ 0.190	2.295 $\pm$ 0.088	2.158 $\pm$ 0.083	1.703 $\pm$ 0.068
Non-differentiable	200	1.773 $\pm$ 0.031	2.010 $\pm$ 0.154	1.598 $\pm$ 0.053	1.478 $\pm$ 0.062	1.042 $\pm$ 0.024
Non-differentiable	500	1.159 $\pm$ 0.017	1.252 $\pm$ 0.070	0.9773 $\pm$ 0.0239	0.8805 $\pm$ 0.0256	0.5052 $\pm$ 0.0108
3-D diffusion	50	1.230 $\pm$ 0.244	2.324 $\pm$ 0.643	0.4641 $\pm$ 0.0487	0.4838 $\pm$ 0.0683	0.2341 $\pm$ 0.0272
3-D diffusion	100	0.8385 $\pm$ 0.1177	1.547 $\pm$ 0.526	0.2087 $\pm$ 0.0080	0.2184 $\pm$ 0.0138	0.1045 $\pm$ 0.0108
3-D diffusion	200	0.8315 $\pm$ 0.1139	0.9617 $\pm$ 0.4736	0.08337 $\pm$ 0.00500	0.09950 $\pm$ 0.00610	0.04064 $\pm$ 0.00191
3-D diffusion	500	0.5716 $\pm$ 0.0531	0.4896 $\pm$ 0.1593	0.03189 $\pm$ 0.00139	0.03625 $\pm$ 0.00130	0.01426 $\pm$ 0.00039
3-D diffusion	1000	0.4956 $\pm$ 0.0372	0.2787 $\pm$ 0.0560	0.01503 $\pm$ 0.00077	0.01755 $\pm$ 0.00074	0.00752 $\pm$ 0.00013

MAPE is inapplicable to the signed planar fields and is therefore absent, rather than represented by empty table cells.

3.4 Curve-level comparison and ablations

Every cell reports the arithmetic mean and sample standard deviation (ddof = 1) across ten paired seeds. For each function, a fixed-reference efficiency ratio uses the maximum canonical budget:

$$\text{efficiency ratio} = \frac{N_{\max}}{\min\{N : \overline{\text{IAE}}_{\text{CETV}}(N) \leq \overline{\text{IAE}}_{\text{reference}}(N_{\max})\}}.$$

No interpolation is used. The twofold test requires a ratio of at least two against both SMART and gSMART and a CETV mean at least one corresponding reference sample standard deviation below both means at a strict majority of canonical budgets. We report this operational test separately from budget-by-budget mean dominance rather than using one to imply the other.

A paired mechanism control replaces CETV’s selected structural teacher by the generic LOO/IDW fallback at $N = 250$ on each planar field. On diffusion, the matched $N = 500$ control is SMART, which removes both the physical-response polynomial and its response-measure weighting. All controls retain the initial design, evaluation set, ordinary-Kriging reporter, and sequential protocol.

3.5 Software and execution

The implementation uses Python 3.11, NumPy, SciPy, scikit-learn, Matplotlib, and the supplied gpCAM 8.4.1 checkout [Harris et al., 2020, Virtanen et al., 2020, Hunter, 2007, Noack et al., 2022]. gpCAM supplies covariance assembly, exact factorization, rank updates, and candidate ranking; the ordinary-mean correction and acquisition callables are in the submitted wrapper. Dense held-out prediction is chunked to avoid a quadratic test covariance. Bluesky’s orchestration model [Allan et al., 2019] and SciAnalysis’s diffraction reduction [Yager, 2025] are deployment context: the synthetic response already corresponds to the scalar quantity their beamline loop would measure and reduce.

4 Experimental Results

4.1 Qualitative reproduction of the adaptive baselines

The definitive baseline curves satisfy the qualitative conditions needed for an interpretable comparison. SMART and gSMART mean IAE decreases strictly at every successive canonical checkpoint on all five fields. On three-dimensional diffusion, both adaptive policies beat both grid and random at every

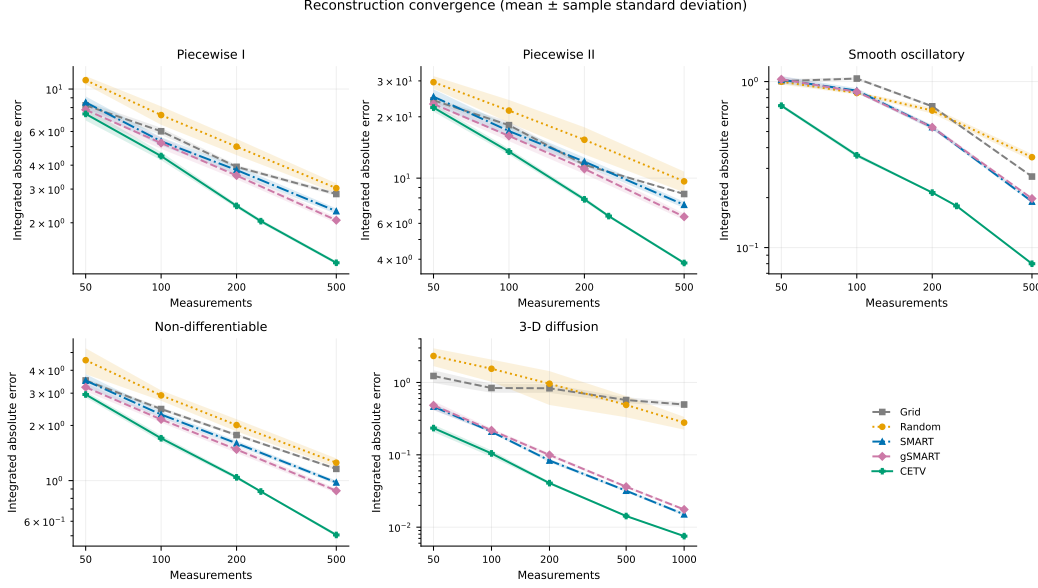


Figure 1: **Response-scale reconstruction error.** Curves show mean integrated absolute error and shaded sample standard deviation over ten paired seeds. Every policy uses the same ordinary-Kriging reporter. The exact planar $N = 250$ CETV points test the half-budget threshold; they are not interpolated values.

budget. At $N = 1000$, SMART and gSMART reach 0.01503 ± 0.00077 and 0.01755 ± 0.00074 , compared with 0.4956 ± 0.0372 for grid and 0.2787 ± 0.0560 for random. On the smooth oscillatory field, final gSMART error is 0.1974 ± 0.0027 , below grid (0.2679 ± 0.0021) and random (0.3504 ± 0.0161). Thus the earlier pathologies of static sampling dominating every diffusion checkpoint and final smooth-field gSMART regressing behind both static policies are absent.

Finite early exceptions remain visible in Table 1. At $N = 50$, SMART is slightly worse than grid on both piecewise fields and both adaptive policies are slightly worse than the static policies on the smooth field. At $N = 200$, SMART remains slightly above grid on Piecewise II. By $N = 500$, both adaptive policies beat both static policies on all four planar fields, and gSMART improves over SMART on Piecewise I, Piecewise II, and the non-differentiable map. These exceptions are retained because qualitative reproduction concerns the full trend and limiting ordering, not selective deletion of early cells.

4.2 Four exact twofold crossings

CETV is below both adaptive means at every matched canonical budget on every field. More importantly, four curves achieve the stronger baseline’s final mean at the exact half budget. On Piecewise I, CETV at $N = 250$ is 2.0394 ± 0.0470 , below final gSMART (2.0626 ± 0.0722). Smooth oscillation reaches 0.17821 ± 0.00204 , below final SMART (0.18922 ± 0.00376). The non-differentiable field reaches 0.87201 ± 0.00900 , below final gSMART (0.88046 ± 0.02560). Finally, diffusion at $N = 500$ reaches 0.014258 ± 0.000394 , below final SMART (0.015028 ± 0.000770). The corresponding half-budget margins against the stronger references are 1.13%, 5.82%, 0.96%, and 5.12%.

The second condition is curve-level rather than a favorable-checkpoint count. CETV is at least one reference standard deviation below both adaptive means at three of four canonical budgets on Piecewise I, all four budgets on the smooth and non-differentiable fields, and all five diffusion budgets. It therefore demonstrates strict twofold sample efficiency on four of five functions under the predeclared operational definition.

The same ordering persists beyond the crossing. Relative to the stronger final adaptive baseline, CETV’s final mean reductions are 40.1% on Piecewise I, 57.8% on smooth oscillation, 42.6% on

Table 2: **Observed crossings of maximum-budget adaptive accuracy.** Crossing N is the first evaluated CETV budget whose mean is no larger than the reference mean at its maximum canonical budget. The classification additionally requires one-reference-standard-deviation separation from both adaptive means at a majority of canonical budgets.

Function	Reference	crossing N	efficiency ratio	classification
Piecewise I	SMART	250	$2.00\times$	substantial
Piecewise I	gSMART	250	$2.00\times$	substantial
Piecewise II	SMART	250	$2.00\times$	mixed or insufficient
Piecewise II	gSMART	500	$1.00\times$	
Smooth oscillatory	SMART	250	$2.00\times$	substantial
Smooth oscillatory	gSMART	250	$2.00\times$	substantial
Non-differentiable	SMART	250	$2.00\times$	substantial
Non-differentiable	gSMART	250	$2.00\times$	substantial
3-D diffusion	SMART	500	$2.00\times$	substantial
3-D diffusion	gSMART	500	$2.00\times$	substantial

the non-differentiable field, and 50.0% on diffusion. Since headline IAE and the separately stored ordinary-Kriging IAE agree exactly in every trajectory, these gains arise from the acquired locations rather than a policy-specific reporting model.

4.3 The Piecewise-II near-miss

Piecewise II is the sole field that does not satisfy the joint criterion. CETV at $N = 250$ gives 6.5140 ± 0.1108 . This is below final SMART (7.3963 ± 0.2609) but 0.0491 IAE, or 0.76%, above final gSMART (6.4649 ± 0.2390). No interpolation or uncertainty overlap is used to convert that miss into a crossing. The classification is therefore mixed or insufficient in Table 2.

The negative label should not be confused with a directionless result. CETV is more than one reference standard deviation below both adaptive means at all four canonical budgets and is lower in every matched-budget comparison. At $N = 500$, it reaches 3.8352 ± 0.0635 , 40.7% below final gSMART. The structural reason for the threshold miss is examined in Section 5.2.

4.4 Predictive accuracy and reliability

The IAE improvements coincide with lower maximum-budget RMSE and CRPS on all five fields. For example, smooth-field CETV reaches RMSE 0.09598 ± 0.00102 and CRPS 0.05835 ± 0.00088 , compared with SMART's 0.2446 ± 0.0052 and 0.1469 ± 0.0034 . On diffusion the CETV values are 0.2743 ± 0.0370 and 0.2626 ± 0.0186 , compared with SMART's 0.6869 ± 0.0592 and 0.3610 ± 0.0213 . Complete values for all 105 canonical method configurations are reported in Tables 4 and 5.

Calibration does not inherit the point-error ordering. At maximum budget, CETV's mean absolute coverage error exceeds gSMART's on Piecewise I, Piecewise II, and the non-differentiable field. The stationary continuous posterior remains misspecified at a jump even when sampling reduces its mean error. The diffusion posterior is also strongly conservative at low nominal levels. Its final CETV MAPE is $1.1372 \pm 0.0675\%$, worse than SMART ($0.6926 \pm 0.0327\%$) and gSMART ($0.5085 \pm 0.0228\%$) despite better IAE, RMSE, and CRPS. MAPE weights the low-diffusion tail while CETV targets physical absolute loss. For this reason, percentage error is reported only in the fully populated three-dimensional table, and every probability-coverage vector is preserved rather than represented solely by a scalar summary.

Posterior calibration at the maximum canonical budget

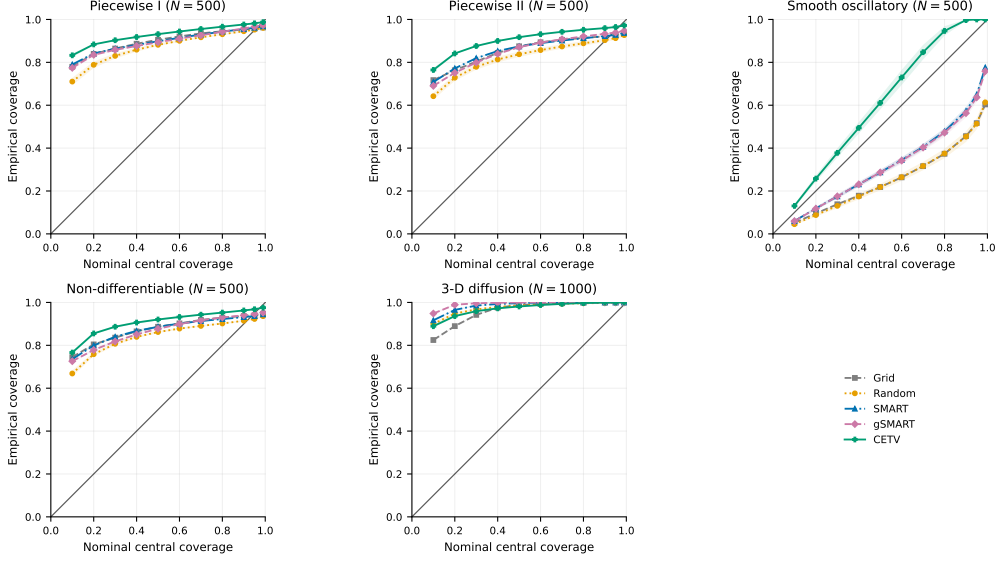


Figure 2: **Maximum-budget probability coverage.** Curves are ten-seed mean empirical coverage of central predictive intervals. The diagonal is ideal. The complete set of curves for every method, function, and budget appears in Figures 5 to 9.

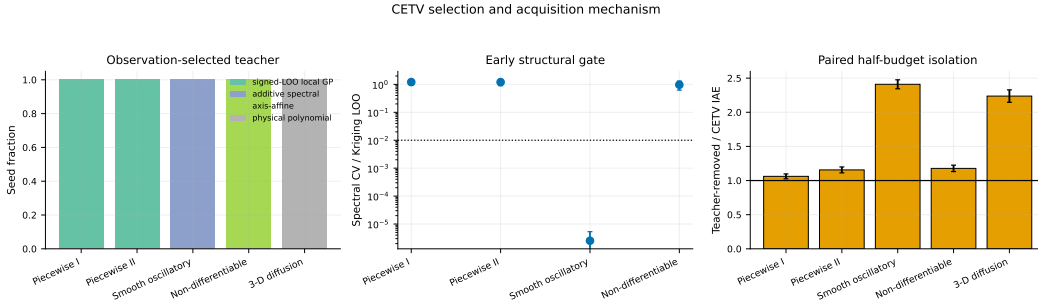


Figure 3: **Teacher selection and paired acquisition isolation.** Left: the selected family is identical across all ten seeds of each field. Middle: the early spectral cross-validation ratio, with the fixed 0.01 gate. Right: paired half-budget IAE after replacing the selected teacher by the generic LOO/IDW error-volume control in two dimensions or SMART in three; values above one favor CETV. Error bars are sample standard deviations.

5 Analysis

5.1 Mechanism: validated mismatch becomes acquisition error

The central distinction between CETV and variance sampling is causal. Posterior variance measures uncertainty inside the reporter’s covariance family. A teacher disagreement measures a failure of that family that has already been supported by observed responses. Multiplication by $r^{d\zeta}$ then approximates how much unresolved measure can carry a similar error. The score is high only when three conditions coincide: the teacher passes an observed-data structural test, it disagrees with ordinary Kriging at a candidate, and enough local volume remains unsampled.

The observation-only gates behave consistently rather than acting as unstable switches. All ten Piecewise-I and Piecewise-II trajectories select the signed-LOO local-GP teacher. Their mean spectral ratios are 1.212 ± 0.269 and 1.206 ± 0.303 , and their normalized axis-affine training errors are 0.1013 ± 0.0233 and 0.0849 ± 0.0260 , far above their respective 0.01 and 0.02 admission thresholds.

Table 3: **Paired structural-teacher removals.** Planar controls use the generic cross-validated LOO/IDW error volume at $N = 250$; the diffusion control is SMART at $N = 500$. Ratios summarize seedwise control IAE divided by CETV IAE.

Function	retained CETV teacher	removed component	CETV	ablation	paired ablation/CETV
Piecewise I	signed-LOO local GP	selected structural teacher	2.039 ± 0.047	2.164 ± 0.053	1.062 ± 0.036
Piecewise II	signed-LOO local GP	selected structural teacher	6.514 ± 0.111	7.527 ± 0.224	1.156 ± 0.043
Smooth oscillatory	additive spectral	selected structural teacher	0.1782 ± 0.0020	0.4295 ± 0.0128	2.410 ± 0.066
Non-differentiable	axis-affine	selected structural teacher	0.8720 ± 0.0090	1.027 ± 0.034	1.178 ± 0.047
3-D diffusion	physical polynomial	physical response teacher	0.01426 ± 0.00039	0.03189 ± 0.00139	2.237 ± 0.091

Smooth oscillation selects the additive spectral teacher in every seed with a ratio $(2.52 \pm 2.81) \times 10^{-6}$. The non-differentiable field fails the spectral gate but admits the axis-affine teacher with normalized training error 0.00116 ± 0.00209 . Diffusion uses the declared positive-response polynomial branch. Function identity is absent from all of these decisions.

The removals in Table 3 isolate the improvement from the unchanged reporter. Replacing the selected teacher raises paired mean IAE by factors 1.062 ± 0.036 on Piecewise I, 1.156 ± 0.043 on Piecewise II, 2.410 ± 0.066 on smooth oscillation, and 1.178 ± 0.047 on the non-differentiable map. Removing the physical polynomial and measure weighting in favor of SMART raises diffusion error by 2.237 ± 0.091 . These are new sequential trajectories, not retrospective substitutions into one fixed design.

The families succeed for different, interpretable reasons. The spectral teacher identifies the analytic field’s common oscillation frequency from only observed folds, so direct Kriging-to-teacher discrepancy targets phase error that variance treats as stationary fill distance. The axis-affine teacher places response changes into four simple regions and lets the fixed geometry-power cycle alternate interface refinement with coverage. The signed-LOO partition uses the sign of reporter error to form local experts, preventing large residuals on opposite sides of a jump from canceling. Finally, the diffusion teacher estimates error in the physical response and multiplies it by the exact log-radius integration Jacobian. The common principle is validated reporter mismatch; the teacher supplies the structural coordinate in which that mismatch is predictable.

5.2 Failure analysis: unequal intersecting jumps

Piecewise II contains the same radial and vertical boundary geometries as Piecewise I but assigns much larger and unequal offsets to the three regions. It is neither additively spectral nor nearly represented by a four-leaf axis-affine tree at $N = 40$, so the selector correctly falls through to the signed-LOO local-GP family. That teacher reduces error consistently, as the 1.156 removal ratio and every matched-budget ordering demonstrate. Its failure is more specific: a local residual partition must locate both intersecting boundaries while a stationary ordinary-Kriging reporter spreads an error proportional to the jump amplitude across the wrong side of each boundary. At 250 measurements, the remaining boundary-location bias is large enough to miss gSMART’s 500-measurement mean by 0.0491 IAE.

This diagnosis explains why simply increasing exploitation is not a remedy. The unsuccessful development record includes direct edge and jump teachers, larger local partitions, denser candidate scoring, residual-process GPs, integrated variance reduction, tree ensembles, and geometry powers from 0.5 through 2.5. Several improve one boundary while degrading the other. The retained policy is therefore not presented as a universal structural oracle: Piecewise II shows that the teacher library can recognize misspecification yet still lack a representation accurate enough for a twofold crossing.

Uncertainty reliability supplies a second failure mode. CETV optimizes absolute map error, not posterior calibration, and its locations do not change the stationary Gaussian likelihood. Near discontinuities, central intervals are generally conservative over most of the domain while remaining locally overconfident at the jump. Better IAE and CRPS can therefore coexist with worse mean absolute coverage error. A noise-aware calibration layer would be a distinct modeling contribution and is deliberately not folded into the acquisition comparison.

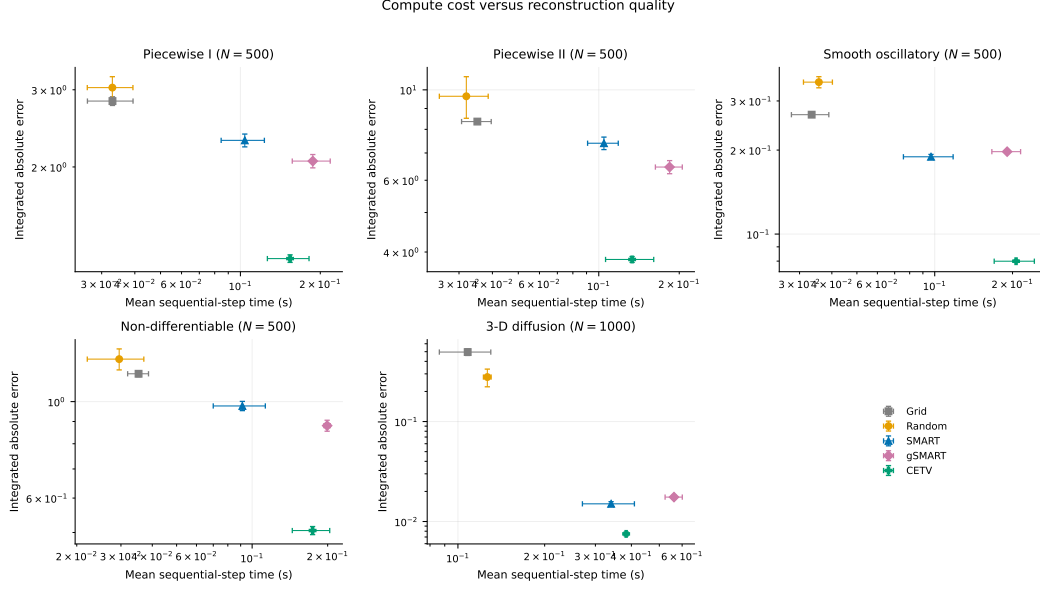


Figure 4: **Maximum-budget error versus mean sequential-step time.** Both axes are logarithmic. Points and error bars are ten-seed means and sample standard deviations. Held-out checkpoint prediction is excluded from step time for every policy.

5.3 Computational cost trade-off

CETV pays for cross-validation, teacher refits, and disagreement evaluation. On Piecewise I, Piecewise II, and the non-differentiable map, its mean step times are 0.154, 0.133, and 0.174 s, between SMART (0.104, 0.105, 0.091) and gSMART (0.187, 0.184, 0.199) s. Diffusion similarly costs 0.384 s per step, compared with 0.340 for SMART and 0.562 for gSMART. Smooth oscillation is the expensive exception: scoring the complete independent bank raises CETV to 0.206 s, slightly above gSMART’s 0.191 s and more than twice SMART’s 0.097 s.

Figure 4 and Table 6 places this overhead beside final error. Complete CETV trajectories average 66.7 to 102.8 s on the planar fields and 391.8 s on diffusion; these totals include common held-out checkpoint evaluation. Subsecond decision cost is likely small relative to motor motion, detector exposure, and diffraction reduction, but no facility latency was measured here. The defensible conclusion is algorithmic: CETV’s four twofold sample-efficiency results require modestly more compute than variance ranking and are usually no more expensive than gradient ranking.

5.4 Scope, sensitivity, and limitations

The study contains five deterministic fields, ten initial designs, and only dimensions two and three. It does not test counting noise, drift, heteroscedastic reduction uncertainty, inaccessible sample regions, or motion cost. The structural library is intentionally small, but it is also well matched to three benchmark regularities: additive periodicity, axis-affine regions, and a smooth physical trend. A response outside those families may fall through to the local teacher, as Piecewise II does, or may require a new teacher whose admission threshold must be validated independently.

The geometry-power cycle is a fixed sensitivity choice. On the non-differentiable field, the best single fixed power reached 0.88171 at half budget, whereas the frozen four-step cycle reaches 0.87201 ± 0.00900 . This modest difference is sufficient for the strict crossing and cautions against treating the selected schedule as a universal constant. Likewise, the spectral phase’s exhaustive ranking is an optimizer choice coupled to that acquisition family, although it sees the same independent candidate bank and no additional responses.

Method development also included component-confirmation sweeps on the same ten initial-design seeds later used for the headline trajectories. The reported standard deviations therefore quantify

sequential-design variability conditional on a selected method; they are not an independent estimate of algorithm-selection uncertainty. A stronger external test would freeze CETV and evaluate it on new response maps, new initial designs, and a physical beamline without revisiting the gates.

Finally, the finite candidate maximizer is not a certificate of the continuous maximum, and the stationary exponential reporter is both a fair common control and a source of interface bias. Nonstationary or physics-informed reporting could reduce that bias, but any such extension must be ablated separately from sampling. These limits define the next scientific test: retain observation-only teacher selection while adding noise-aware errors, motion cost, and an independently validated reporter.

6 Conclusion

Cross-validated error-teacher volume converts model mismatch into a measurement decision without replacing the model being evaluated. Early observed-data gates select an additive spectral, axis-affine, signed-LOO local-GP, or physical-response polynomial teacher. The teacher’s disagreement with ordinary Kriging is calibrated by held-out residuals and weighted by unresolved response-map volume. All gates and fitting rules are fixed across fields, and unmeasured responses never enter the selector.

Under a common exponential-kernel ordinary-Kriging reporter, CETV reaches the stronger adaptive policy’s maximum-budget mean error at exactly half the measurements on Piecewise I, smooth oscillation, the non-differentiable map, and three-dimensional diffusion. It also satisfies the required one-standard-deviation separation on a majority of each curve. Piecewise II is the honest exception: its half-budget mean is 0.76% above final gSMART, although CETV is better at every matched budget and ends 40.7% lower. Thus the principal result is a strict acquisition-only twofold improvement on four of five response maps, not a universal claim.

The corrected reference policies are themselves qualitatively sound. SMART and gSMART improve monotonically on every field, both dominate static sampling at every diffusion budget, and final smooth-field gSMART beats both grid and random. Paired teacher removals increase half-budget error on all five fields, while the unchanged reporter makes the acquisition contribution identifiable. RMSE and CRPS improve with IAE, but complete reliability curves show that uncertainty near discontinuities remains conservative and that diffusion percentage error favors a different region of response space.

These results identify both the value and the boundary of error teachers. When observed data support a simple structural explanation for reporter failure, sampling that discrepancy can halve measurement cost. When two large unequal interfaces must be localized by a generic local committee, the same policy may yield strong uniform dominance without crossing the exact twofold threshold. Future work should test a frozen teacher library on unseen maps and noisy physical closed loops, while preserving the isolation used here: gains in a response map must remain attributable to where the experiment measured.

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A Supplementary Evidence and Reproducibility

A.1 Trajectory and metric integrity

Every summary is generated from serialized sequential trajectories. The strict analysis validates 250 headline records from five fields, five policies, and ten seeds, plus 40 planar LOO/IDW teacher-removal controls. The diffusion removal uses the SMART records already present in the headline product, giving 290 distinct validated records. Before writing a table or figure, the analysis enforces the following invariants:

1. Policies compared for one field and seed have identical ten-point initial designs and independent candidate banks, verified from regenerated coordinates and SHA-256 hashes.
2. The held-out design is independent of acquisition, shared across policies and seeds for a field, and has no exact overlap with any measured point.
3. Each checkpoint is an exact prefix of one sequential trajectory. There is no restart, curve interpolation, failed-run imputation, or retrospective point selection.
4. Every stored measurement response is re-evaluated from the benchmark function, and the lengths of the response, variogram-fit, acquisition-fit, and timing histories agree with the requested budget.
5. Headline IAE equals separately stored ordinary-Kriging IAE in every record. CETV states explicitly that teacher selection does not use a function label, candidate truth, or held-out response.
6. IAE, RMSE, CRPS, every coverage vector, and diffusion MAPE are finite at every canonical checkpoint.

The complete audit, seedwise values, gate fits, speedup classifications, and artifact list are in `proposal/logs/cetv_aggregate.json`. Source trajectories are in `proposal/logs/trajectories/definitive` and `proposal/logs/trajectories/cetv_main`. CETV launch completion is recorded in `proposal/logs/cetv_main_2d_manifest.json` and `proposal/logs/cetv_main_3d_manifest.json`.

A.2 Complete predictive-score matrices

Planar and volumetric metrics are separated because percentage error is meaningful only for the strictly positive diffusion coefficient. RMSE uses the predictive mean, CRPS scores the complete Gaussian or lognormal distribution [Gneiting and Raftery, 2007], and MACE is the mean absolute difference between nominal and empirical central-interval coverage over the eleven levels in Section 3.3. Calibration and sharpness are distinct forecast properties [Gneiting et al., 2007, Kuleshov et al., 2018]. The planar table therefore has no inapplicable MAPE column. The three-dimensional table contains a finite MAPE value in every row.

Table 4: Planar predictive metrics at every canonical checkpoint. RMSE and CRPS are mean $\pm$ sample standard deviation over ten seeds; MACE summarizes the ten-seed mean coverage curve. Lower is better.

Function	N	Method	RMSE	CRPS	MACE
Piecewise I	50	Grid	17.86 ± 1.10	7.442 ± 0.297	0.194
Piecewise I	50	Random	22.84 ± 2.06	9.469 ± 0.562	0.114
Piecewise I	50	SMART	20.47 ± 1.70	7.875 ± 0.530	0.195
Piecewise I	50	gSMART	18.40 ± 1.59	7.125 ± 0.444	0.187
Piecewise I	50	CETV	16.02 ± 1.47	6.736 ± 0.450	0.193
Piecewise I	100	Grid	17.91 ± 0.85	6.261 ± 0.250	0.265
Piecewise I	100	Random	18.27 ± 2.08	6.762 ± 0.632	0.197
Piecewise I	100	SMART	15.39 ± 0.85	5.484 ± 0.221	0.257
Piecewise I	100	gSMART	14.97 ± 1.21	5.236 ± 0.311	0.238
Piecewise I	100	CETV	11.65 ± 0.93	4.567 ± 0.143	0.246
Piecewise I	200	Grid	14.37 ± 0.37	4.320 ± 0.082	0.287
Piecewise I	200	Random	15.25 ± 1.61	5.086 ± 0.441	0.255
Piecewise I	200	SMART	13.32 ± 1.09	4.235 ± 0.193	0.289
Piecewise I	200	gSMART	12.06 ± 0.50	3.892 ± 0.120	0.273
Piecewise I	200	CETV	7.807 ± 0.226	3.406 ± 0.115	0.315
Piecewise I	500	Grid	12.38 ± 0.28	3.503 ± 0.059	0.324
Piecewise I	500	Random	11.59 ± 0.76	3.481 ± 0.160	0.300
Piecewise I	500	SMART	10.25 ± 0.45	2.917 ± 0.075	0.323
Piecewise I	500	gSMART	9.281 ± 0.511	2.673 ± 0.071	0.319
Piecewise I	500	CETV	5.333 ± 0.121	2.313 ± 0.043	0.350
Piecewise II	50	Grid	46.67 ± 1.38	21.05 ± 0.52	0.150
Piecewise II	50	Random	56.49 ± 5.60	25.47 ± 2.09	0.122
Piecewise II	50	SMART	50.66 ± 4.83	22.30 ± 2.02	0.157
Piecewise II	50	gSMART	46.44 ± 2.68	20.15 ± 0.89	0.135
Piecewise II	50	CETV	43.45 ± 2.06	19.47 ± 0.79	0.152
Piecewise II	100	Grid	44.72 ± 2.08	17.47 ± 0.73	0.216
Piecewise II	100	Random	47.47 ± 7.13	19.47 ± 2.58	0.175
Piecewise II	100	SMART	40.94 ± 2.86	16.18 ± 1.00	0.208
Piecewise II	100	gSMART	38.31 ± 3.00	15.02 ± 0.83	0.185
Piecewise II	100	CETV	31.20 ± 1.58	13.14 ± 0.40	0.205
Piecewise II	200	Grid	35.56 ± 0.64	11.97 ± 0.18	0.252
Piecewise II	200	Random	39.57 ± 5.78	14.84 ± 2.09	0.222
Piecewise II	200	SMART	34.72 ± 2.02	12.26 ± 0.56	0.247
Piecewise II	200	gSMART	31.85 ± 2.29	11.21 ± 0.59	0.229
Piecewise II	200	CETV	22.73 ± 0.77	9.261 ± 0.259	0.269
Piecewise II	500	Grid	30.78 ± 0.22	9.357 ± 0.054	0.292
Piecewise II	500	Random	30.69 ± 2.94	10.11 ± 0.98	0.266
Piecewise II	500	SMART	27.10 ± 1.07	8.360 ± 0.236	0.293
Piecewise II	500	gSMART	24.38 ± 1.23	7.440 ± 0.228	0.287
Piecewise II	500	CETV	15.46 ± 0.41	6.291 ± 0.089	0.329
Smooth oscillatory	50	Grid	1.241 ± 0.034	0.8845 ± 0.0307	0.434
Smooth oscillatory	50	Random	1.248 ± 0.056	0.8802 ± 0.0503	0.437
Smooth oscillatory	50	SMART	1.262 ± 0.065	0.8880 ± 0.0495	0.420
Smooth oscillatory	50	gSMART	1.270 ± 0.057	0.8979 ± 0.0461	0.425
Smooth oscillatory	50	CETV	0.9444 ± 0.0175	0.5711 ± 0.0139	0.257
Smooth oscillatory	100	Grid	1.307 ± 0.021	0.9521 ± 0.0202	0.479
Smooth oscillatory	100	Random	1.084 ± 0.023	0.7580 ± 0.0126	0.441
Smooth oscillatory	100	SMART	1.098 ± 0.039	0.7856 ± 0.0280	0.449
Smooth oscillatory	100	gSMART	1.086 ± 0.015	0.7737 ± 0.0156	0.447
Smooth oscillatory	100	CETV	0.4541 ± 0.0206	0.2685 ± 0.0094	0.194
Smooth oscillatory	200	Grid	0.9712 ± 0.0204	0.6472 ± 0.0145	0.451
Smooth oscillatory	200	Random	0.8867 ± 0.0328	0.5914 ± 0.0274	0.413
Smooth oscillatory	200	SMART	0.6814 ± 0.0206	0.4583 ± 0.0146	0.389
Smooth oscillatory	200	gSMART	0.6860 ± 0.0244	0.4576 ± 0.0171	0.386
Smooth oscillatory	200	CETV	0.2552 ± 0.0023	0.1530 ± 0.0019	0.131
Smooth oscillatory	500	Grid	0.3695 ± 0.0027	0.2277 ± 0.0020	0.294
Smooth oscillatory	500	Random	0.5106 ± 0.0292	0.2969 ± 0.0137	0.295
Smooth oscillatory	500	SMART	0.2446 ± 0.0052	0.1469 ± 0.0034	0.213
Smooth oscillatory	500	gSMART	0.2587 ± 0.0043	0.1548 ± 0.0027	0.218
Smooth oscillatory	500	CETV	0.09598 ± 0.00102	0.05835 ± 0.00088	0.087
Non-differentiable	50	Grid	7.218 ± 0.278	3.140 ± 0.111	0.163

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Function	N	Method	RMSE	CRPS	MACE
Non-differentiable	50	Random	8.254 ± 1.418	3.827 ± 0.662	0.115
Non-differentiable	50	SMART	7.227 ± 0.467	3.132 ± 0.245	0.162
Non-differentiable	50	gSMART	6.655 ± 0.324	2.852 ± 0.099	0.147
Non-differentiable	50	CETV	5.739 ± 0.214	2.593 ± 0.092	0.150
Non-differentiable	100	Grid	6.169 ± 0.203	2.381 ± 0.069	0.223
Non-differentiable	100	Random	6.304 ± 0.400	2.634 ± 0.131	0.177
Non-differentiable	100	SMART	5.601 ± 0.248	2.211 ± 0.081	0.221
Non-differentiable	100	gSMART	5.058 ± 0.293	2.030 ± 0.080	0.195
Non-differentiable	100	CETV	3.999 ± 0.128	1.704 ± 0.047	0.217
Non-differentiable	200	Grid	5.659 ± 0.091	1.892 ± 0.030	0.275
Non-differentiable	200	Random	5.167 ± 0.414	1.963 ± 0.146	0.233
Non-differentiable	200	SMART	4.650 ± 0.210	1.671 ± 0.052	0.261
Non-differentiable	200	gSMART	4.272 ± 0.284	1.529 ± 0.058	0.244
Non-differentiable	200	CETV	3.025 ± 0.086	1.257 ± 0.023	0.274
Non-differentiable	500	Grid	4.666 ± 0.062	1.347 ± 0.016	0.307
Non-differentiable	500	Random	4.097 ± 0.230	1.362 ± 0.067	0.282
Non-differentiable	500	SMART	3.607 ± 0.131	1.149 ± 0.024	0.305
Non-differentiable	500	gSMART	3.335 ± 0.136	1.054 ± 0.028	0.298
Non-differentiable	500	CETV	2.017 ± 0.067	0.8582 ± 0.0151	0.333

Table 5: Three-dimensional diffusion predictive metrics at every canonical checkpoint. RMSE, CRPS, and MAPE are mean $\pm$ sample standard deviation over ten seeds; MACE summarizes the mean coverage curve. MAPE is in percent. Lower is better.

Function	N	Method	RMSE	CRPS	MACE	MAPE (%)
3-D diffusion	50	Grid	10.89 ± 1.60	1.203 ± 0.114	0.267	11.01 ± 0.61
3-D diffusion	50	Random	15.65 ± 2.63	1.924 ± 0.529	0.243	27.54 ± 16.63
3-D diffusion	50	SMART	5.644 ± 0.385	0.9401 ± 0.0307	0.276	11.20 ± 1.38
3-D diffusion	50	gSMART	5.913 ± 0.871	0.9680 ± 0.0675	0.303	9.278 ± 0.633
3-D diffusion	50	CETV	2.260 ± 0.373	0.6875 ± 0.0586	0.269	29.54 ± 19.71
3-D diffusion	100	Grid	7.056 ± 0.817	0.8541 ± 0.0523	0.298	6.897 ± 0.253
3-D diffusion	100	Random	12.67 ± 2.63	1.330 ± 0.400	0.294	12.67 ± 6.15
3-D diffusion	100	SMART	3.293 ± 0.212	0.7133 ± 0.0275	0.323	6.511 ± 0.548
3-D diffusion	100	gSMART	3.541 ± 0.215	0.7633 ± 0.0283	0.352	4.918 ± 0.489
3-D diffusion	100	CETV	1.225 ± 0.140	0.4951 ± 0.0291	0.309	11.65 ± 4.78
3-D diffusion	200	Grid	6.668 ± 0.725	0.7881 ± 0.0510	0.293	6.360 ± 0.275
3-D diffusion	200	Random	9.073 ± 3.406	0.8912 ± 0.3358	0.334	6.590 ± 1.717
3-D diffusion	200	SMART	1.813 ± 0.209	0.5454 ± 0.0350	0.360	3.327 ± 0.266
3-D diffusion	200	gSMART	2.318 ± 0.158	0.6210 ± 0.0150	0.381	2.510 ± 0.213
3-D diffusion	200	CETV	0.7427 ± 0.1229	0.4178 ± 0.0431	0.346	4.186 ± 0.665
3-D diffusion	500	Grid	5.150 ± 0.400	0.5784 ± 0.0229	0.358	2.588 ± 0.072
3-D diffusion	500	Random	6.022 ± 1.737	0.5673 ± 0.1020	0.375	2.451 ± 0.428
3-D diffusion	500	SMART	0.9900 ± 0.0712	0.4176 ± 0.0190	0.388	1.404 ± 0.102
3-D diffusion	500	gSMART	1.349 ± 0.095	0.5014 ± 0.0176	0.401	1.029 ± 0.055
3-D diffusion	500	CETV	0.4363 ± 0.0665	0.3319 ± 0.0282	0.377	1.982 ± 0.292
3-D diffusion	1000	Grid	4.888 ± 0.290	0.5727 ± 0.0155	0.379	1.829 ± 0.051
3-D diffusion	1000	Random	3.910 ± 0.886	0.4362 ± 0.0328	0.392	1.213 ± 0.108
3-D diffusion	1000	SMART	0.6869 ± 0.0592	0.3610 ± 0.0213	0.401	0.6926 ± 0.0327
3-D diffusion	1000	gSMART	0.9702 ± 0.0756	0.4395 ± 0.0186	0.408	0.5085 ± 0.0228
3-D diffusion	1000	CETV	0.2743 ± 0.0370	0.2626 ± 0.0186	0.389	1.137 ± 0.068

A.3 Probability coverage at every budget

The complete empirical vectors underlying MACE are serialized in `proposal/tables/coverage.csv`, with one row per field, method, budget, and nominal level. Figures 5 to 9 plot every required method/configuration curve. Lines are ten-seed means, shaded bands are sample standard deviations, and the diagonal denotes ideal central coverage.

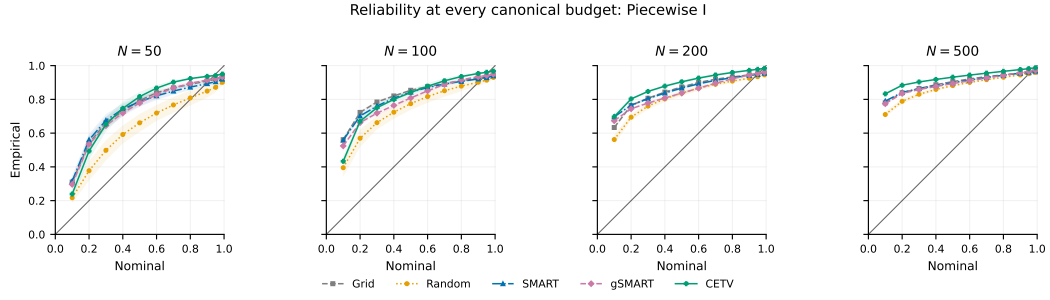


Figure 5: Probability-coverage curves for all five policies at every canonical Piecewise-I budget.

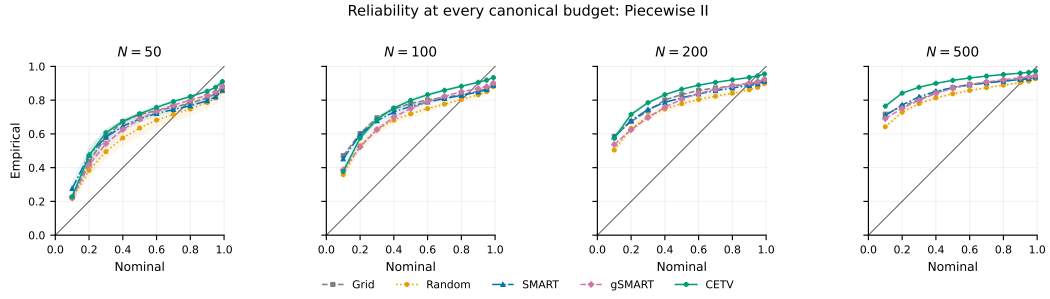


Figure 6: Probability-coverage curves for all five policies at every canonical Piecewise-II budget.

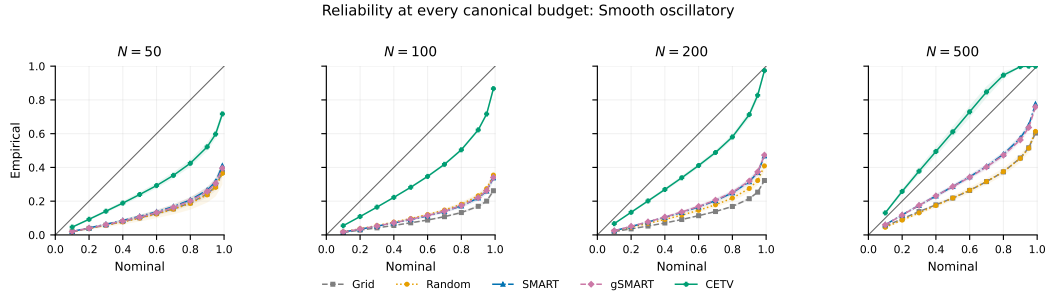


Figure 7: Probability-coverage curves for all five policies at every canonical smooth-oscillation budget.

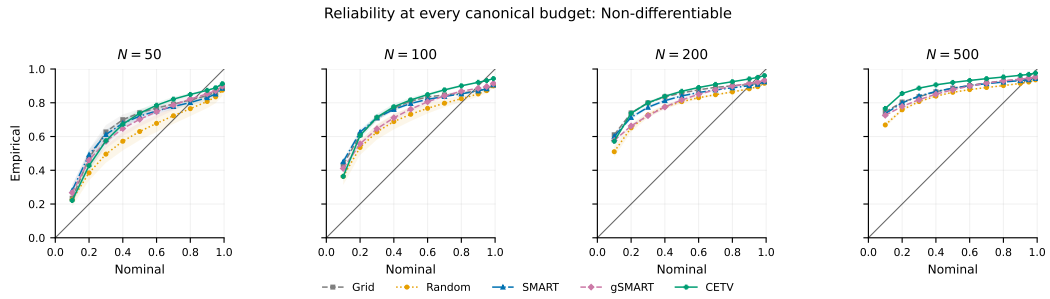


Figure 8: Probability-coverage curves for all five policies at every canonical non-differentiable-map budget.

A.4 Runtime and measurement geometry

Table 6: Runtime accounting. Trajectory time includes sequential acquisition and held-out checkpoint prediction. Mean step time covers the acquisition, response evaluation, and Kriging-refit loop and excludes checkpoint prediction. Entries are mean $\pm$ sample standard deviation over ten seeds.

Function	Method	trajectory time (s)	mean step time (s)
Piecewise I	Grid	17.69 ± 3.46	0.0329 ± 0.0064
Piecewise I	Random	17.67 ± 3.46	0.0329 ± 0.0064
Piecewise I	SMART	52.42 ± 9.74	0.1039 ± 0.0193
Piecewise I	gSMART	93.34 ± 15.22	0.1874 ± 0.0306
Piecewise I	CETV	76.86 ± 13.67	0.1539 ± 0.0275
Piecewise II	Grid	18.80 ± 2.39	0.0351 ± 0.0045
Piecewise II	Random	17.09 ± 3.56	0.0320 ± 0.0066
Piecewise II	SMART	52.73 ± 6.91	0.1046 ± 0.0137
Piecewise II	gSMART	91.67 ± 10.54	0.1841 ± 0.0212
Piecewise II	CETV	66.68 ± 13.56	0.1333 ± 0.0273
Smooth oscillatory	Grid	17.87 ± 2.90	0.0334 ± 0.0055
Smooth oscillatory	Random	19.03 ± 2.44	0.0356 ± 0.0046
Smooth oscillatory	SMART	48.86 ± 10.65	0.0967 ± 0.0211
Smooth oscillatory	gSMART	94.93 ± 12.07	0.1906 ± 0.0243
Smooth oscillatory	CETV	102.81 ± 18.07	0.2064 ± 0.0365
Non-differentiable	Grid	18.86 ± 1.80	0.0353 ± 0.0034
Non-differentiable	Random	15.75 ± 4.00	0.0295 ± 0.0075
Non-differentiable	SMART	46.08 ± 10.77	0.0914 ± 0.0214
Non-differentiable	gSMART	99.11 ± 2.80	0.1990 ± 0.0057
Non-differentiable	CETV	86.72 ± 14.63	0.1742 ± 0.0296
3-D diffusion	Grid	122.11 ± 24.79	0.1081 ± 0.0219
3-D diffusion	Random	142.76 ± 3.84	0.1265 ± 0.0039
3-D diffusion	SMART	351.96 ± 72.31	0.3401 ± 0.0697
3-D diffusion	gSMART	569.12 ± 37.74	0.5622 ± 0.0386
3-D diffusion	CETV	391.81 ± 5.72	0.3836 ± 0.0059

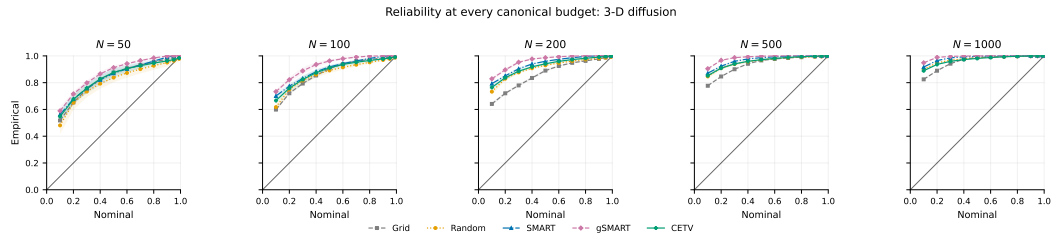


Figure 9: Probability-coverage curves for all five policies at every canonical three-dimensional diffusion budget.

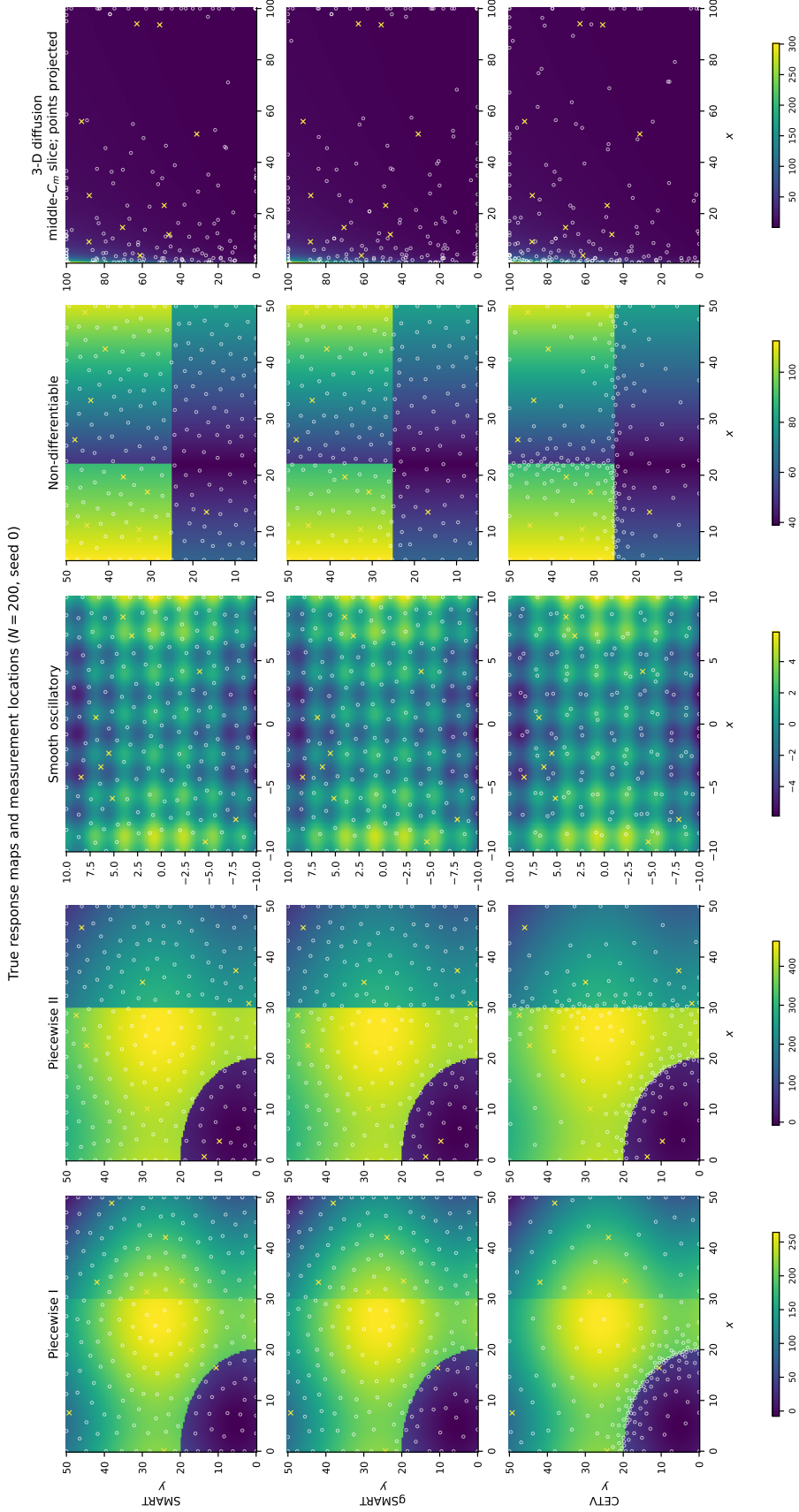


Figure 10: **Representative measurement locations.** Locations at $N = 200$ for seed 0 under SMART, gSMART, and CETV on all five fields. Background color is the true response, yellow crosses are the shared initial design, and open white circles are subsequent acquisitions. Diffusion locations are projected over the middle concentration slice. The fixed seed is used only for visualization; every quantitative comparison averages ten seeds.

A.5 Development record

The chronological record in `proposal/attempts_log.md` documents 44 protocol corrections, candidate criteria, sensitivity sweeps, retained components, and negative results. Rejected approaches include direct edge and jump reconciliation, residual-process GPs, integrated-variance criteria, alternative volume powers, staged hybrids, dense local partitions, tree ensembles, an auxiliary-task error model, and a truth-exposed oracle used only as an explicitly ineligible capacity diagnostic. It also records one resource-invalid 3 GB estimator launch and its compact replacement. No development or oracle record is merged into the headline aggregate.